
Master Thesis

FYSM64

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1 Introduction

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2 Method

3 Result

4 Discussion

5 Linblad Approach

6 Calculation the Quantum Limit

We write the full Hamiltonian as

$$H = H_{\text{det}} + H_{\text{sys}} + H_{\text{int}} + H_{\text{env}}, \quad (6.1)$$

where H_{det} is the Hamiltonian of the detector defined as

$$H_{\text{det}} = \omega_r a^\dagger a + \frac{\omega_q}{2} \sigma_z + g(a^\dagger \sigma_- + a \sigma_+), \quad (6.2)$$

H_{sys} is the Hamiltonian of the measured system which is defined, for a DQD in energy basis, as

$$H_{\text{sys}} = \frac{\omega_d}{2} \tau_z, \quad (6.3)$$

H_{int} is the interaction Hamiltonian between the detector and the measured system, defined as

$$H_{\text{int}} = g_d [C \tau_z \sigma_z + S(\tau_+ \sigma_- + \tau_- \sigma_+)], \quad (6.4)$$

and H_{env} is the Hamiltonian consisting of the environment causing relaxation and dephasing, described as

$$H_{\text{env}} = H_{\text{cavity}} + H_{\text{rel}}^\sigma + H_\phi^\sigma + H_{\text{rel}}^\tau + H_\phi^\tau. \quad (6.5)$$

The cavity Hamiltonian is defined as

$$H_{\text{cavity}} = \int d\omega \omega b^\dagger(\omega) b(\omega) + i \sqrt{\frac{\kappa}{2\pi}} \int d\omega [b^\dagger(\omega) a - b(\omega) a^\dagger], \quad (6.6)$$

the relaxation Hamiltonians for the qubits are defined similarly but with Γ_1^τ and Γ_1^σ instead of κ and their own bosonic operators, $\xi_{\phi/\sigma}$. The dephasing Hamiltonians are defined by

$$H_\phi^\sigma = \int d\omega \omega \xi_\phi^\dagger(\omega) \xi_\phi(\omega) + \sigma_z \int d\omega \lambda_\omega [\xi_\phi^\dagger(\omega) + \xi_\phi(\omega)] \quad (6.7)$$

We define

$$H_{\text{closed}} = H_{\text{det}} + H_{\text{sys}} + H_{\text{int}}. \quad (6.8)$$

Using Heisenberg's equation of motion we obtain

$$\dot{b}(\omega) = -i[b(\omega), H] = -i[b(\omega), H_{\text{env}}] = -i\omega b(\omega) + \sqrt{\frac{\kappa}{2\pi}} a \quad (6.9)$$

with solution

$$b(\omega, t) = e^{-i\omega(t-t_0)}b(\omega, t_0) + \sqrt{\frac{\kappa}{2\pi}} \int_{t_0}^t dt' e^{-i\omega(t-t')} a(t'). \quad (6.10)$$

We also define the quantum Langevin equation for an operator o as

$$\dot{o} = -i[o, H] + \sum_l \left[-[o, a_l^\dagger] \left(\frac{\kappa_l}{2} a_l + \sqrt{\kappa} b_{\text{in},l}(t) \right) + \left(\frac{\kappa}{2} a_l^\dagger + \sqrt{\kappa_l} b_{\text{in},l}^\dagger(t) \right) [o, a_l] \right], \quad (6.11)$$

where the H is the Hamiltonian minus the bath and $l \in \{c, i\}$ for c being the coupling port to the transmission line and i being the internal losses. It is also noted that $a_c = a_i = a$, $b_{\text{in},c} = b_{\text{in}}$, $b_{\text{in},i} = \xi$, and $\kappa = \kappa_c + \kappa_i$. Applying the quantum Langevin equation to the system we get the following equations of motion

$$\dot{a} = -\left(i\omega_r + \frac{\kappa}{2}\right)a - ig\sigma_- - \sqrt{\kappa_c}b_{\text{in}}(t) - \sqrt{\kappa_i}\xi(t) \quad (6.12a)$$

$$\dot{\sigma}_- = -(i\omega_q + 2ig_d C\tau_z + \Gamma_2^\sigma)\sigma_- + i(ga + g_d S\tau_-)\sigma_z \quad (6.12b)$$

$$\dot{\tau}_- = -(i\omega_d + 2ig_d C\sigma_z + \Gamma_2^\tau)\tau_- + ig_d S\sigma_- \tau_z \quad (6.12c)$$

$$\dot{\sigma}_z = 2ig(a^\dagger \sigma_- - a\sigma_+) + 2ig_d S(\tau_+ \sigma_- - \tau_- \sigma_+) - \Gamma_1^\sigma(\sigma_z + 1) \quad (6.12d)$$

$$\dot{\tau}_z = 2ig_d S(\sigma_+ \tau_- - \sigma_- \tau_+) - \Gamma_1^\tau(\tau_z + 1), \quad (6.12e)$$

where we have

$$\Gamma_2 = \frac{\Gamma_1}{2} + \Gamma_\phi. \quad (6.13)$$

References

- [1] A. Tavakoli and G. Cobucci, “Introduction to quantum information theory”, Edited by Oxford University Press and in publication, 2026.